

STUDY GUIDE

CONCEPT SUMMARY

Big Ideas

- Any number that can be written as the fraction $\frac{m}{n}$, $n \neq 0$, where m and n are integers, is rational.
- Exponents can be used to represent roots and reciprocals of rational numbers.
- The exponent laws can be extended to include powers with rational and variable bases, and rational exponents.

Applying the Big Ideas

This means that:

- If a real number can be expressed as a terminating or repeating decimal, it is rational; otherwise, it is irrational.
- The numerator of a rational exponent indicates a power, while the denominator indicates a root. A negative exponent indicates a reciprocal.
- We can use the exponent laws to simplify expressions that involve rational exponents.

Reflect on the Chapter

- How can you predict whether the value of a radical will be a rational number or an irrational number?
- How were the exponent laws used to create definitions for negative exponents and rational exponents?
- What does it mean to simplify an expression involving radicals or exponents?



THE WORLD OF MATH

Careers: Financial Planner

The financial services industry offers many career opportunities – in banking, insurance, investment firms, as well as private practice. A personal financial planner uses math and technology to explore different ways for a person to invest her or his money. The ability to understand, manipulate, and evaluate algebraic formulas that involve rational exponents is an essential skill.



SKILLS SUMMARY

Skill	Description	Example
Classify numbers. [4.1, 4.2]	To determine whether a number is rational or irrational, write the number in decimal form. - Repeating and terminating decimals are rational. - Non-repeating, non-terminating decimals are irrational.	Rational numbers: $2, 0, -3, 3.75, 0.0\overline{1}, \frac{3}{5}, -\frac{10}{7}$ Irrational numbers: $\sqrt{3}, \pi$
Simplify radicals. [4.3]	To simplify a square root: - Write the radicand as a product of its greatest perfect square factor and another number. - Take the square root of the perfect square factor. A similar procedure applies for cube roots and higher roots.	$\begin{aligned}\sqrt{200} &= \sqrt{100 \cdot 2} \\ &= \sqrt{100} \cdot \sqrt{2} \\ &= 10\sqrt{2}\end{aligned}$ $\begin{aligned}\sqrt[3]{200} &= \sqrt[3]{8 \cdot 25} \\ &= \sqrt[3]{8} \cdot \sqrt[3]{25} \\ &= 2\sqrt[3]{25}\end{aligned}$
Evaluate powers. [4.4, 4.5]	To evaluate powers without using a calculator: - Rewrite a power with a negative exponent as a power with a positive exponent. - Represent powers with fractional exponents as radicals. - Use mental math to evaluate the powers and/or simplify the roots.	$\begin{aligned}64^{-\frac{2}{3}} &= \frac{1}{64^{\frac{2}{3}}} \\ &= \frac{1}{(\sqrt[3]{64})^2} \\ &= \frac{1}{4^2} \\ &= \frac{1}{16}\end{aligned}$
Apply the exponent laws to simplify expressions. [4.6]	To simplify expressions involving powers: - Remove brackets by applying the exponent laws for products of powers, quotients of powers, or powers of powers. - Write the simplest expression using positive exponents.	$\begin{aligned}\left(\frac{(xy^2)^3}{x^5y}\right)^{-4} &= \left(\frac{x^3y^6}{x^5y}\right)^{-4} \\ &= \left(\frac{x^5y^1}{x^3y^6}\right)^4 \\ &= (x^{5-3}y^{1-6})^4 \\ &= (x^2y^{-5})^4 \\ &= x^8y^{-20} \\ &= \frac{x^8}{y^{20}}\end{aligned}$